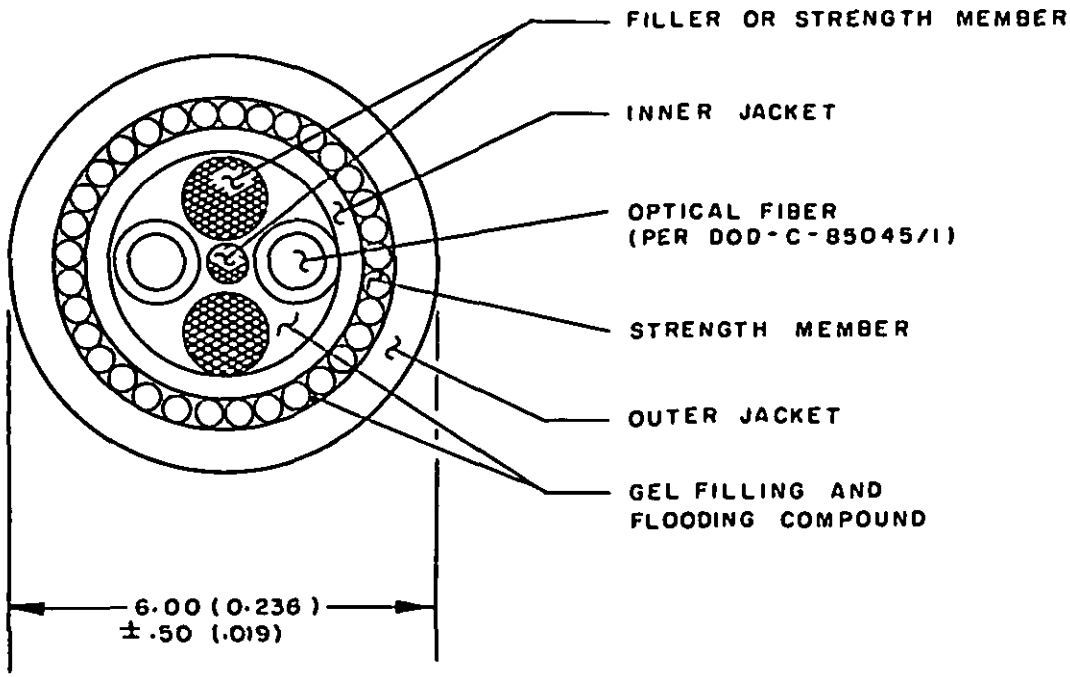
| METRIC |
DOD-C-85045/3A
20 March 1987
SUPERSEDING
DOD-C-85045/3
22 June 1984

MILITARY SPECIFICATION SHEET

CABLES, FIBER OPTICS, HEAVY DUTY, WITH GEL FILLING AND FLOODING COMPOUND (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cables described herein shall consist of this specification and the latest issue of DOD-C-85045.



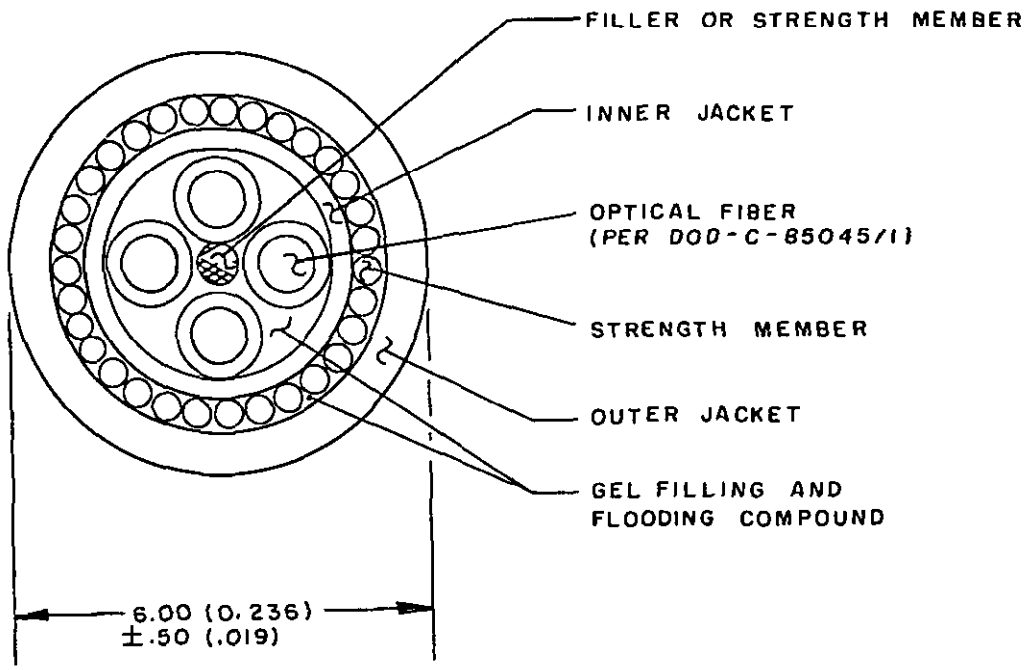
NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are in parentheses.

FIGURE 1. Two fiber cable.

AMSC N/A FSC 6015

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

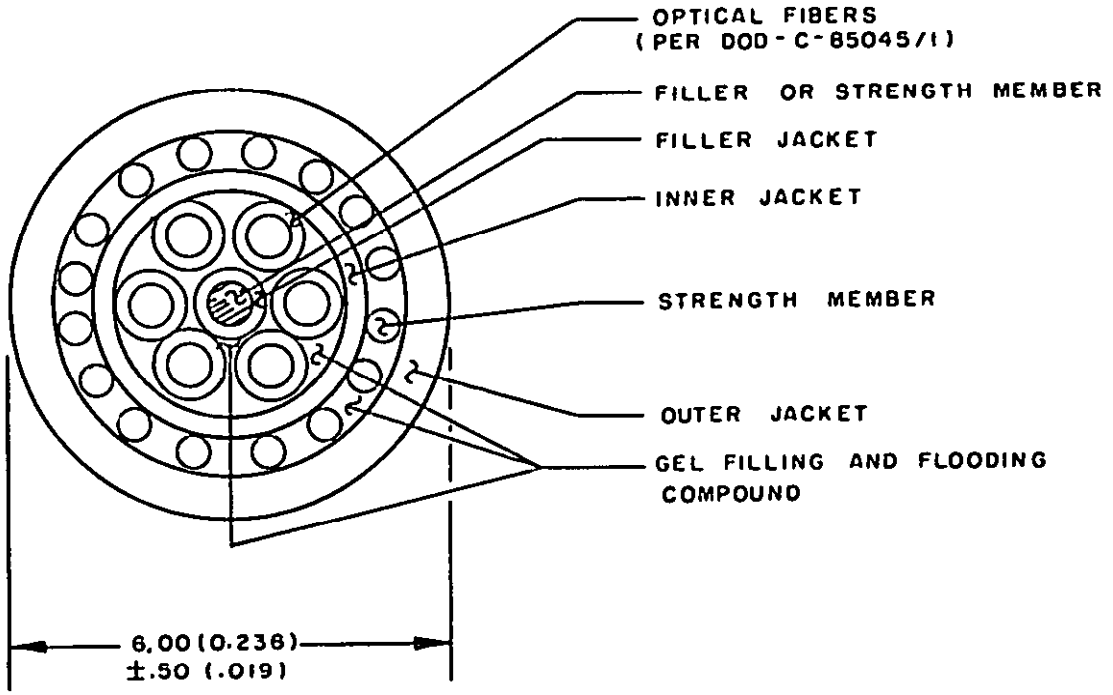
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NOTES:
1. Dimensions are in millmeters.
2. Inch equivalents are in parentheses.

FIGURE 2. Four fiber cable.

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NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are in parentheses.

FIGURE 3. Six fiber cable.

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TABLE I. Part numbers.

Part number D85045/3-	Number of fibers	Maximum weight (kg/km)
B2B	2	25
C2B	2	25
B4B	4	40
C4B	4	40
B6B	6	60
C6B	6	60

REQUIREMENTS:

Dimensions and configuration: Fiber cables shall be designed as in figure 1 through 3.

Weight: See table I.

Fiber: Shall be in accordance with DOD-C-85045/1.

Number of fibers: See table I.

Jacket material: Polyurethane or equivalent.

Color: Outer cable jacket shall be black.

Filler and strength member material: Aramid yarn or equivalent.

Attenuation: Attenuation shall not exceed 3 dB/km at 1100 to 1300 nm or 6 dB/km at 850 ±20 nm for each and every fiber in the cable.

Optical fiber bandwidth (fiber transfer function): The optical fiber bandwidth shall meet or exceed 600 MHz-km at 1300 ±20 nm and 400 MHz-km at 850 ±20 nm.

Numerical aperture: 0.20 ±0.02.

Loose tube buffers: Constructed of thermoplastic polymer capable of containing up to five individual fibers.

Temperature cycling: -46°C and +71°C are the high and low temperature extremes.

Crush: Applicable tests shall be conducted at +25°C ±2°C. The test load of 500 newtons shall be applied within 5 seconds; hold time shall be 10 seconds minimum, release time shall be 5 seconds maximum.

Cable twist - bend: Applicable 2,000 cycles.

Wicking: Procedure II, except that the test duration shall be 168 hours. Maximum dye penetration shall be 5 cm.

Tensile loading: 2,700 newtons minimum.

Storage temperature: The cable shall be stored for 240 hours minimum at -55°C ±2°C and +85°C ±2°C. Following storage at each temperature, the attenuation measured at ambient conditions shall not exceed that specified herein and the exterior surface shall show no cracks or defects.

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Fluid immersion (DOD-STD-1678, method 8030): After testing, the cable diameter shall not exceed the maximum cable diameter specified. Fluids, temperature and immersion time shall be as specified below.

Fluid	Temperature (C°)	Immersion time (hours)
Lubricating oil MIL-L-23699	48 - 50	20
Hydraulic fluid MIL-H-5605	48 - 50	20
Isopropyl alcohol TT-I-735	20 - 25	168
Turbine fuel MIL-T-5624	20 - 25	169
Automobile gas MIL-G-3056	20 - 25	168

Flammability: 60° angle procedure.

Jacket opacity: Shall be verified at 1,300 nm ±50 nm.

Parameter applicability: See table II.

Fiber and strength member motion: Cable utilizes loose tube construction.

Gel filling and flooding compound compatibility: The gel filling and flooding compound shall be compatible with the fiber buffer material.

TABLE II. Parameter applicability. 1/ 2/

Inspection	Specimen quantity and length	Qualification	Quality assurance group A	Quality assurance group B	Quality assurance group C
Group I					
Visual and mechanical	6 units 1 km each	X	X		
Attenuation rate	6 units 1 km each	X	X		
Optical fiber bandwidth	6 units 1 km each	X	X		
Crosstalk	6 units 1 km each	X	X		
Numerical aperture	6 units 2 m each (1 each from 6 cables)	X	X		
Group II					
Temperature cycling	6 units 1 km each (3 on reel, 3 off reel)	X		X	
Humidity	2 units 1 km each	X		X	
Storage temperature	1 units 1 km each (1 on reel, 1 off reel)	X		X	
Barometric pressure (reduced)	1 unit 1 km	X		X	
Tensile loading	6 units 150 m each (2 each from 3 cables)	X		X	
Freezing water immersion (ice crush)	2 units 30 m each (1 each from 2 cables)	X		X	

See footnotes at end of table.

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TABLE II. Parameter applicability - Continued. 1/ 2/

Inspection	Specimen quantity and length	Qualification	Quality assurance group A	Quality assurance group B	Quality assurance group C
Group III (Military environments)					
Low temperature flexibility (cold bend)	3 units 8 m each (1 each from 3 cables)	X			X
Cyclic flexing	6 units 5 m each (3 each from 2 cables, 2 units for each temperature)	X			X
Crush	6 units 5 m each (3 each from 2 cables, 2 units for each temperature)	X			X
Cable twist-bending	6 units 5 m each (3 each from 2 cables, 2 units for each temperature)	X			X
Impact	9 units 5 m each (3 each from 3 cables)	X			X
Jacket self-adhesion or blocking	2 units >5 m each (1 each from 2 cables)	X			X
Wicking	2 units 2 m each (1 each from 2 cables)	X			X
Cable jacket opacity	3 units 2 m each (1 each from 2 cables)	X			X
Fluid immersion	2 units (for each test fluid required) 2 m each (1 each from 2 cables)	X			X
Knot	6 units 5 m each (3 each from 2 cables)	X			X
Cable element removability	3 units 2 m each (1 each from 3 cables)	X			X
Durability of identification marking	3 units 2 m each (1 each from 3 cables)	X			X
Flammability	3 units .6 m each (1 each from 3 cables)	X			X

1/ Tolerance on 1 km lengths is ±2% provided results are normalized to 1 km.

2/ Tolerance on shorter lengths is ±10%.

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Supersession data: See table III.

TABLE III. Supersession data.

Superseding part number DOD-C-85045/3-	Superseded part number DOD-C-85045/3-
B2B	2-C
C2B	2-D
B4B	4-C
C4B	4-D
B6B	6-C
C6B	6-D

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI, SC
Navy - AS
Air Force - 11
DLA - ES

User activity:
Air Force - 17

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6015-0013-2)

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